CoRL 2026 TAG Workshop Oral

5-10 min version

RoboJudge

Human-aligned embodied video judges

Multimodal Language Models as Judges for Embodied Video Generation

8-minute oral: problem, core idea, evidence, and scope.

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800	12K	0.719
benchmark videos	training videos	human Pearson r

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arXiv:2610.27782 | spherelab.ai/RoboJudge | github.com/Sphere-AI-Lab/RoboJudge

Embodied video generation needs a sharper judge

General video quality is not the same as useful robotic rollout quality.

- Generated rollouts can look plausible while violating contact, agent identity, or object dynamics.
- They can also be physically reasonable but miss the instruction or task goal.
- For robotics, the evaluator has to separate physical adherence from instruction alignment.

Physical Adherence

Does the video obey embodied physics and interaction constraints?

Instruction Alignment

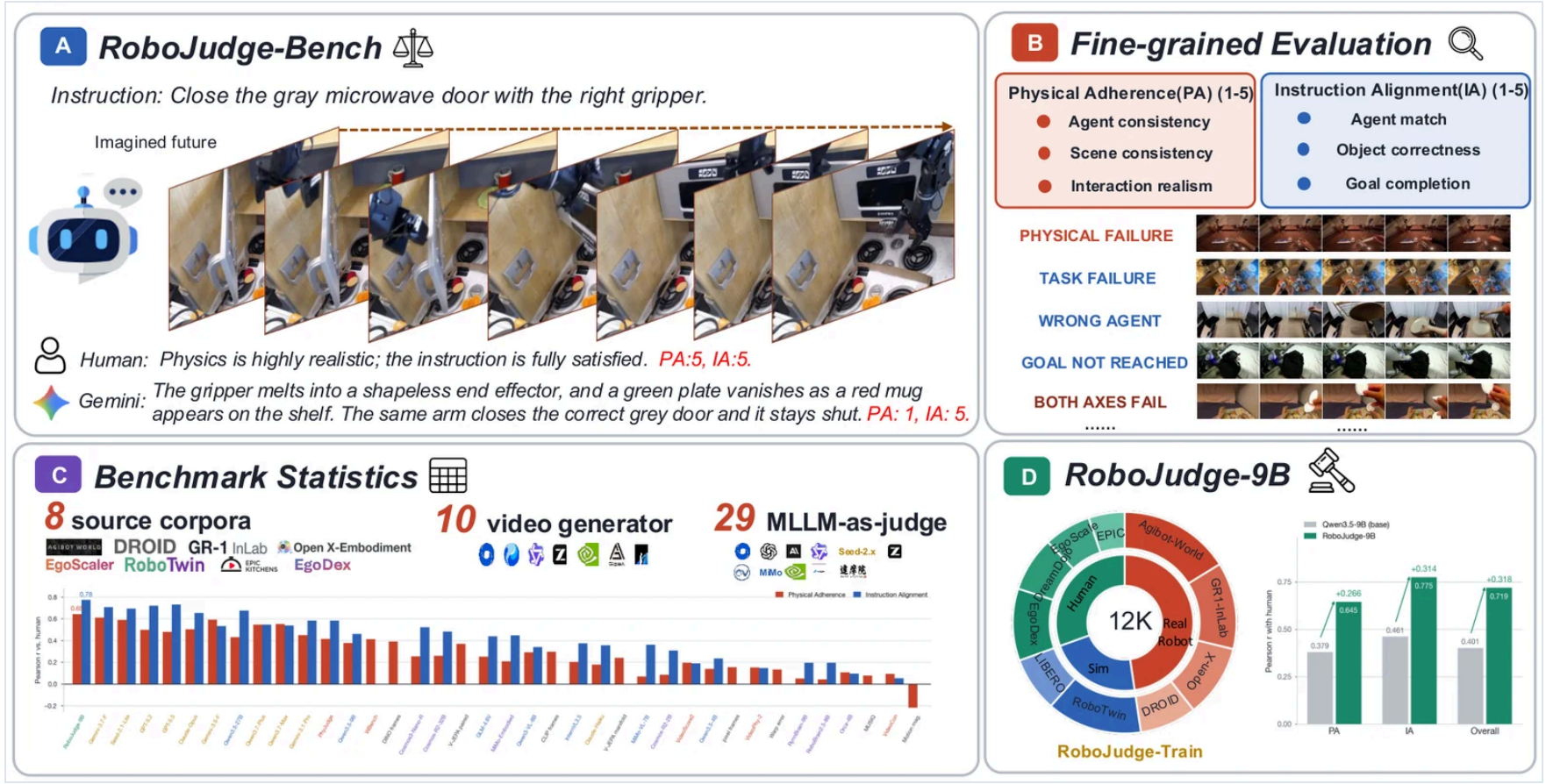
Does the agent, object, and goal match the requested task?

Human agreement

A useful judge should match trained human ratings, not only generic aesthetics.

RoboJudge in one slide

A benchmark, a training set, and a 9B embodied video judge.

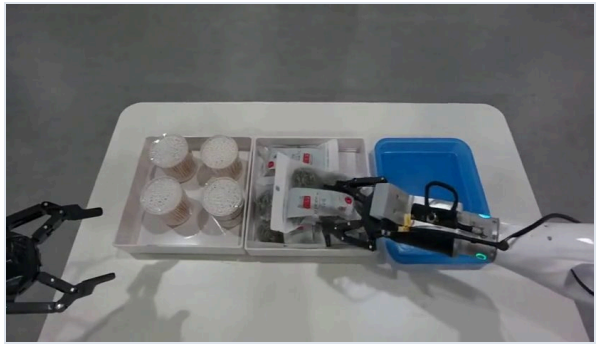


800	12K	29	0.719
human-annotated benchmark videos	training videos with human-verified scores	model judges evaluated	Pearson r with human ratings

The two-axis rubric catches different failure modes

PA and IA are intentionally separated instead of collapsed into one vague score.

Axis	What it checks	Sub-criteria
PA	Physical Adherence	agent consistency; scene consistency; interaction realism
IA	Instruction Alignment	agent match; object correctness; goal completion



AgiBot World / Cosmos3-Nano

PA 5 | IA 3



EPIC-KITCHENS / Wan2.6-Flash

PA 3 | IA 5

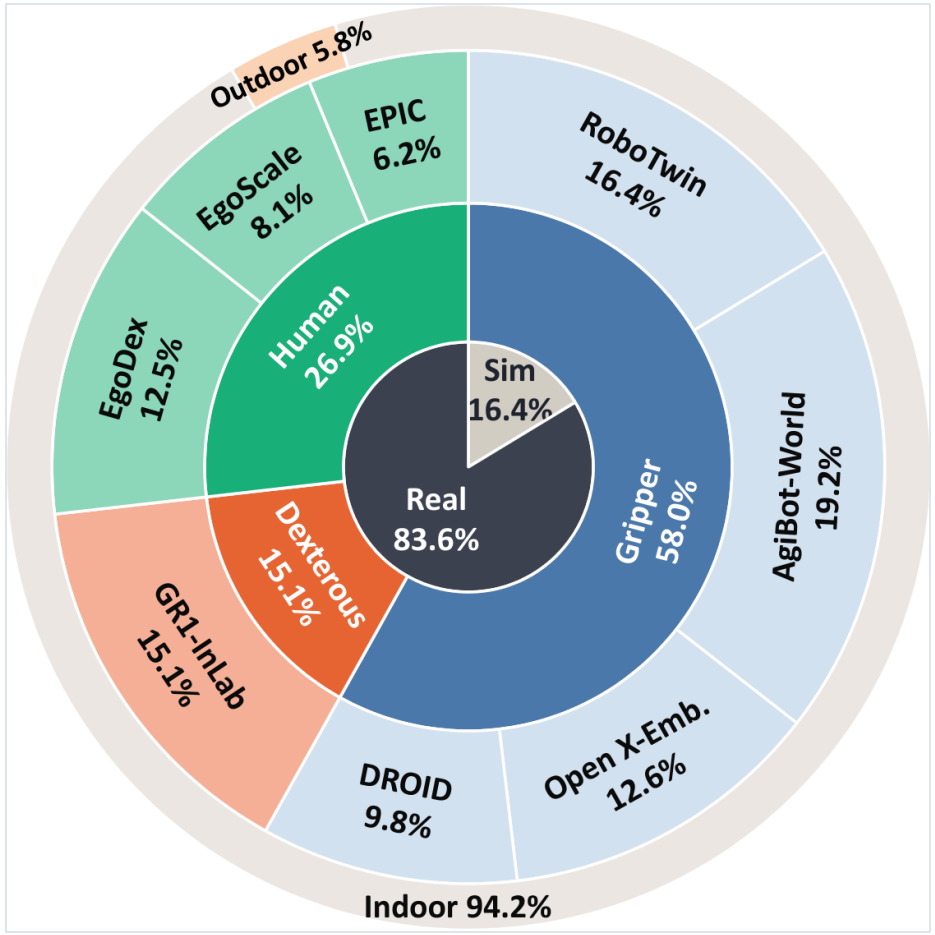


GR-1 InLab / Wan-14B

PA 1 | IA 1

Benchmark coverage is embodied, not generic video

RoboJudge-Bench spans sources, embodiments, generators, and real/sim settings.



8 source corpora

Robot, hand, and human-object interaction video sources.

10 video generators

General, world-model, and embodied generation systems.

Embodiment diversity

Grippers, dexterous humanoids, and human hands.

Sim and real

131 simulated and 669 real-source videos in the benchmark.

Training RoboJudge-9B

Supervised rationale learning followed by score-agreement RL.

RoboJudge-Train

12,351 videos
24,702 axis-level samples

Base

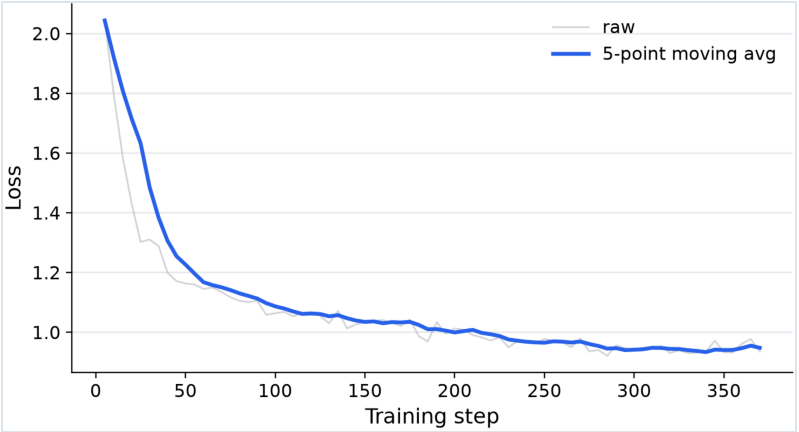
Qwen3.5-9B
video MLLM backbone

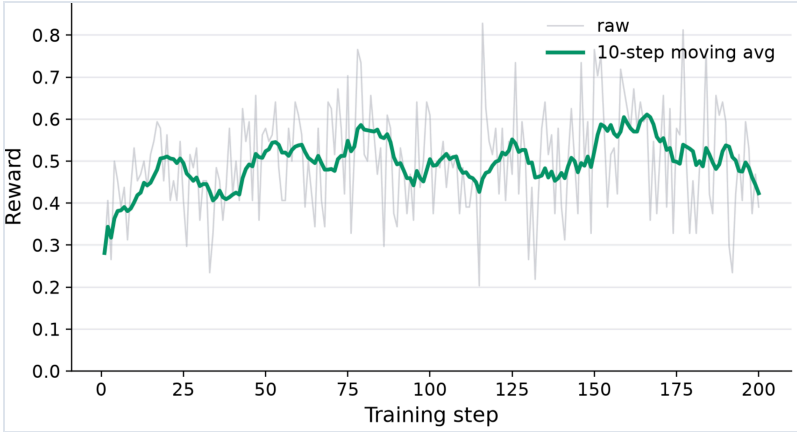
SFT

Learn rubric reasoning
and score format

GRPO RL

Reward exact-score
agreement





RoboJudge-9B agrees best with human ratings

Overall Pearson r on RoboJudge-Bench; higher is better.



+0.089

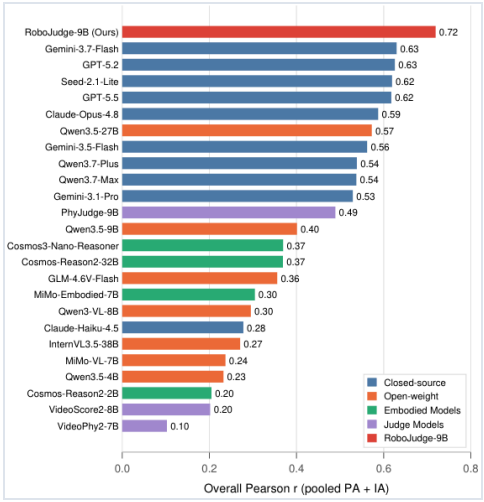
absolute gain over strongest proprietary baseline

0.645

PA Pearson r

0.775

IA Pearson r

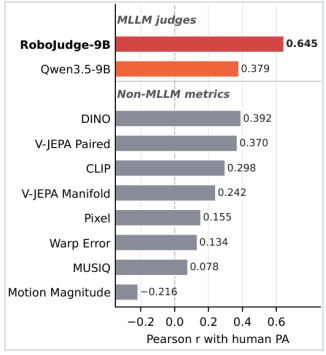


SFT gives the jump; RL sharpens the judge

Ablation on Overall Pearson r, plus out-of-domain checks.



OOD benchmark	Base	+SFT	+SFT+RL
VideoPhy2 PC r	0.179	0.299	0.346
VideoPhy2 SA r	0.412	0.552	0.572
Physion-Eval J	0.056	0.105	0.093

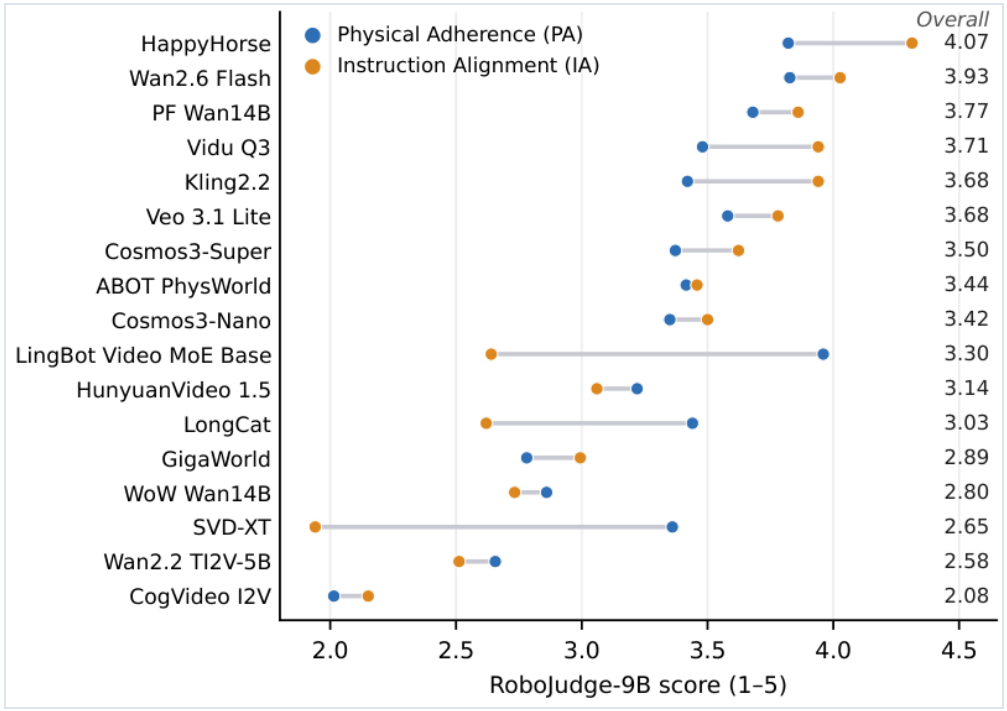


- Specialized video metrics do not transfer cleanly to embodied PA/IA.
- RoboJudge remains a judge of video quality, not a direct robot policy evaluator.

Use case: ranking embodied video generators

RoboJudge-9B can scale PA/IA evaluation across model submissions.

Model	PA	IA	Overall
HappyHorse	3.821	4.313	4.067
Wan2.6 Flash	3.827	4.027	3.927
PF Wan14B	3.680	3.860	3.770
Vidu Q3	3.480	3.940	3.710
Cosmos3-Super	3.372	3.623	3.497
Wan2.2 TI2V-5B	2.656	2.512	2.584
CogVideo I2V	2.014	2.151	2.082



- PA and IA expose different strengths: a model can be more physical than task faithful, or vice versa.
- The leaderboard is an example of scalable evaluation, not a replacement for task-specific robot trials.

Takeaways

What I want you to remember from the short talk.

1. Evaluation target

Embodied video needs PA and IA, not only aesthetics or text-video matching.

2. Evidence

General MLLMs and specialized video judges are not reliably human-aligned here.

3. Result

RoboJudge-9B reaches 0.719 Pearson r and leads the evaluated judges.

Project page: https://spherelab.ai/RoboJudge/

Code: github.com/Sphere-AI-Lab/RoboJudge | Model + dataset: Hugging Face